

BLE Series

Data Sheet

USE_45

BLE 5.3 Wireless Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	03/03/2025	DJ	AB	Initial Release
2.0	25/06/2025	AD	AK	Added pin description
3.0	28/07/2025	DJ	AD	Updated Pin description
4.0	01/08/2025	DJ	AD	Added Assembly recommendation and module placement strategy

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General

The USE_45 is a low-power, highly secure, single-chip wireless module that integrates a high performance Bluetooth Low Energy version 5.3 radio.

For more information reach out to us at info@unisemindia.com.

Features

- DC/DC voltage range: 1.8 V to 3.6 V
- Up to 96 MHz Arm Cortex®-M33 core
- Up to 1 MB flash memory
- 128 KB SRAM
- 8 KB Code Cache to improve performance and efficiency
- Secure Boot ROM
- Programmable Transmit Output Power up to +10 dBm
- Data Rates: 125 kbps, 500 kbps, 1 Mbps, and 2 Mbps
- On board CAN Transceiver
- Typical active core current: <5.3 mA at 96 MHz (<55 μ A/MHz)
- Typical RX: 4.7 mA
- Typical TX at 0 dBm: 4.6 mA and 18.7 mA at 10 dBm

Applications

- Secure Car Access
- Wireless Battery Management Systems (WBMS)
- Positioning/Localization
- Keyless Entry
- Process/Factory Automation
- Building Control and Monitoring

Pinout Diagram :

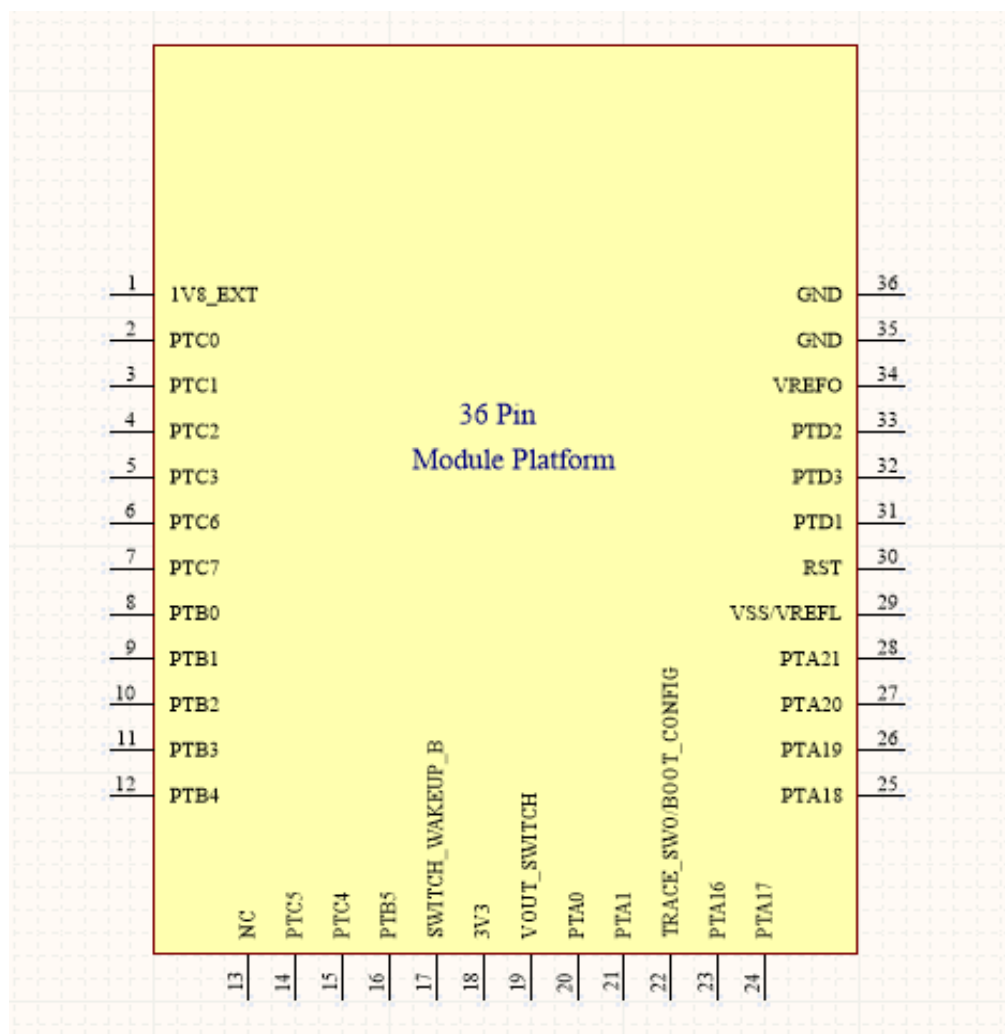


Fig 1 : Module Pinouts

Pin Descriptions:

Pin No.	Pin Name	Type	Function Description
1	1V8_EXT	PI	1.8V External Power input
2	PTC0	I/O	Programmable GPIO.
3	PTC1	I/O	Programmable GPIO.
4	PTC2	I/O	Programmable GPIO.
5	PTC3	I/O	Programmable GPIO.
6	PTC6	I/O	Digital IO/Analog
7	PTC7	I/O	GPIO pin
8	PTB0	I/O	Digital IO/Analog
9	PTB1	I/O	Programmable GPIO/Analog
10	PTB2	I/O	Programmable GPIO/Analog
11	PTB3	I/O	Programmable GPIO/Analog
12	PTB4	I/O	Programmable GPIO.
13	NC	-	No connection
14	PTC5	I/O	Programmable GPIO.
15	PTC4	I/O	Programmable GPIO.
16	PTB5	I/O	GPIO pin
17	SWITCH_WAKEUP_B	Digital/non-GPIO	Internally pulled up to the switch input through a resistor, it can be pulled down to wake up the smart power switch.
18	+3V3	PI	3.3V Power input
19	VOUT_SWITCH	Power	Smart power switch output voltage

20	PTA0	I/O	Programmable GPIO.
21	PTA1	I/O	Programmable GPIO.
22	TRACE_SWO/BOOT_CONFIG	I/O	Boot configuration pin/Analog
23	PTA16	I/O	Programmable GPIO/Analog
24	PTA17	I/O	Programmable GPIO/Analog
25	PTA18	I/O	GPIO pin
26	PTA19	I/O	GPIO pin
27	PTA20	I/O	Digital IO/Analog
28	PTA21	I/O	Digital IO/Analog
29	VSS/VREFL	GND	Ground pin
30	RST	I	Reset pin/Digital IO/Analog
31	PTD1	I/O	Digital IO/Analog
32	PTD3	I/O	Digital IO/Analog
33	PTD2	I/O	Digital IO/Analog
34	VREFO	I/O	Internally generated voltage reference output. When the VREF output buffer is disabled, the VREF_OUT signal state is high-impedance.
35	GND	GND	Ground pin
36	GND	GND	Ground pin

***Please Refer Chapter 20 Signal Muxing and Pinout of KW45 reference manual (Download from <https://www.nxp.com/products/KW45>) Table 20.2 To understand which of this device's signals are multiplexed on which external pin.**

Module Diagram :

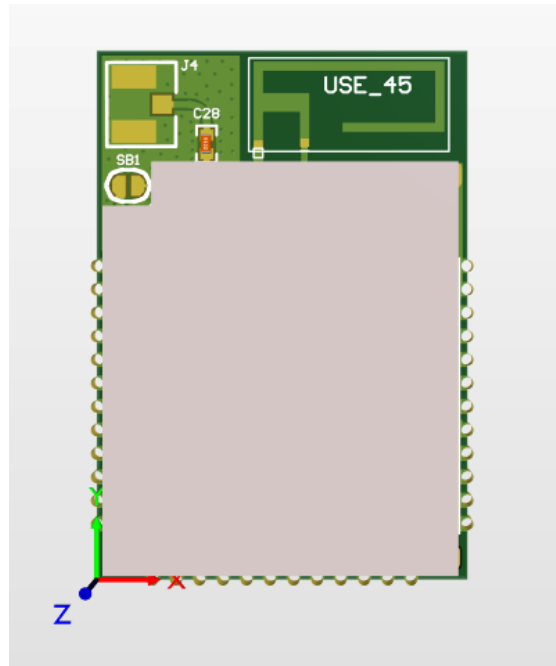


Fig 2(a) : Module (Top view)

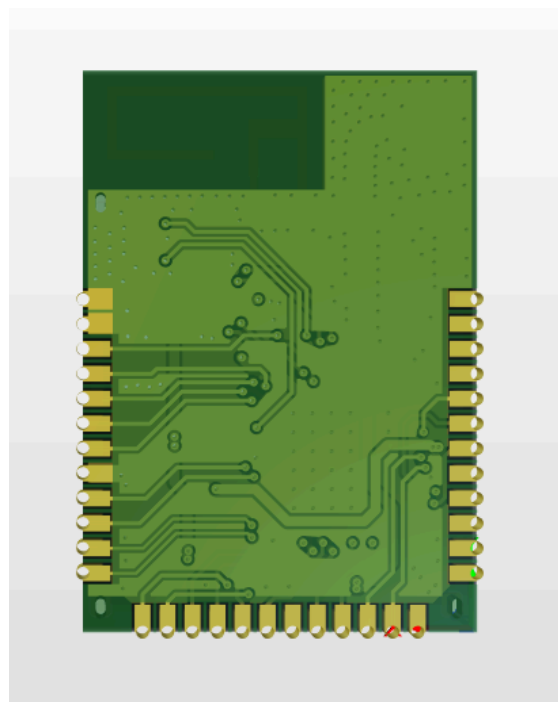


Fig 2(b) : Module (Bottom view)

Module Dimension Drawing:

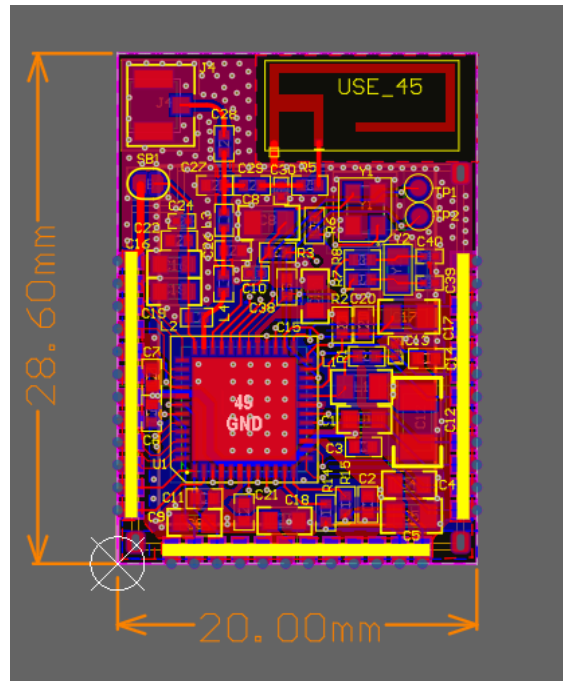


Fig 3

Recommended Footprint Dimension:

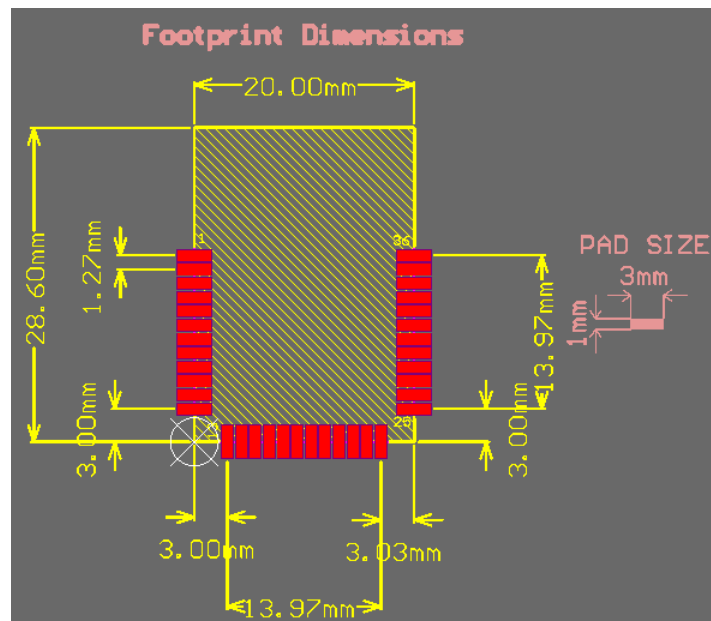


Fig 4

Recommended Footprint Clearance:

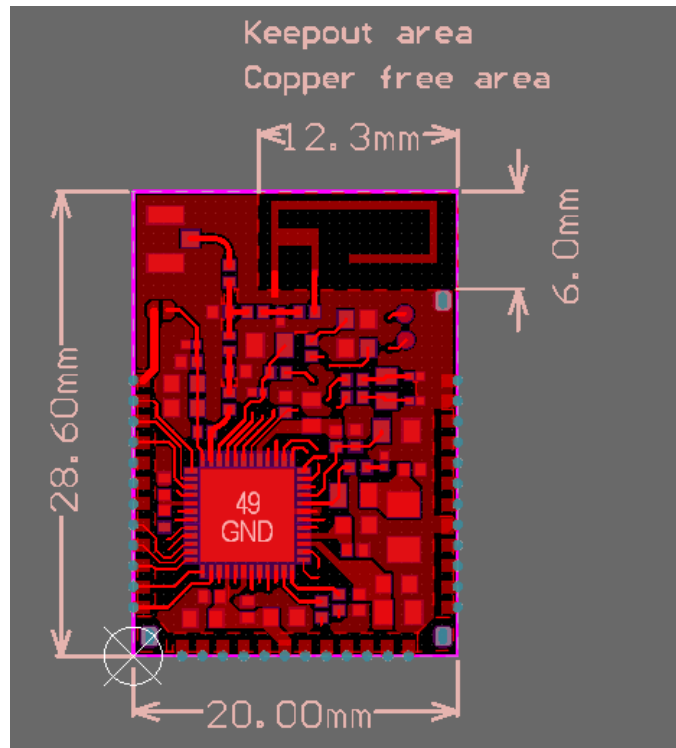


Fig 5

Recommended Connections:

The diagram (shown below Fig 6.) of the module application on external filter parts, when needed, please design in the mainboard, the parts parameter can depend on the actual need to change. And pay attention to the increase and decrease ramping of supply voltage faster than 0.5v/ms;

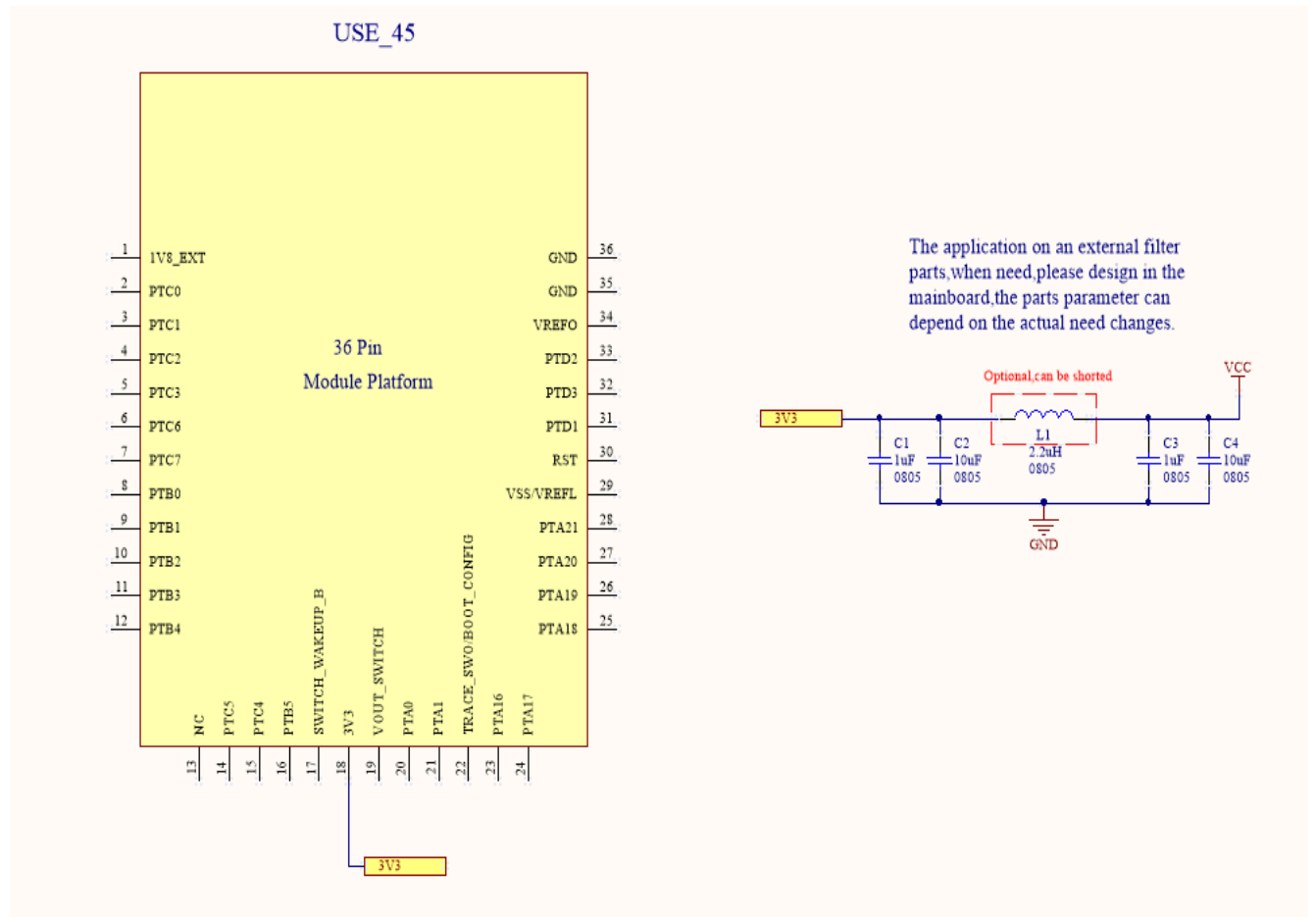


Fig 6

Recommended Assembly:

Signal trace and power supply trace, don't cross layout, as shown in below Fig. To avoid crosstalk, affect the receiving sensitivity.

Assembly recommendation

Underneath the module antenna and RF circuit on the main board PCB copper needs to be cleared, and placed close to the main board edge, as shown in below Fig.

For smaller coin cell batteries, with high worst-case end-of-life equivalent source resistance, a 22- μ F VDDS input capacitor must be used to ensure compliance with this slew rate. Recommended one is to use MLCC for low ESR.

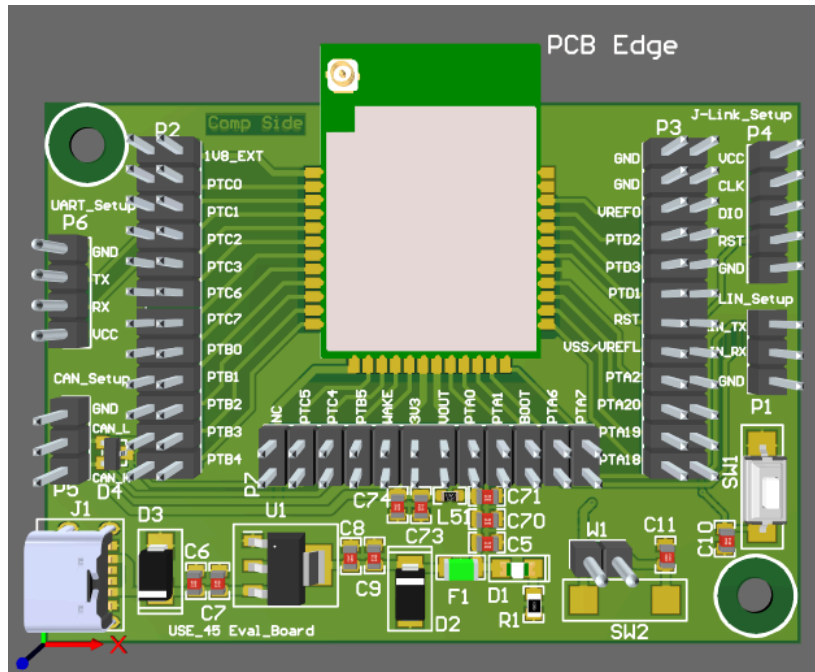
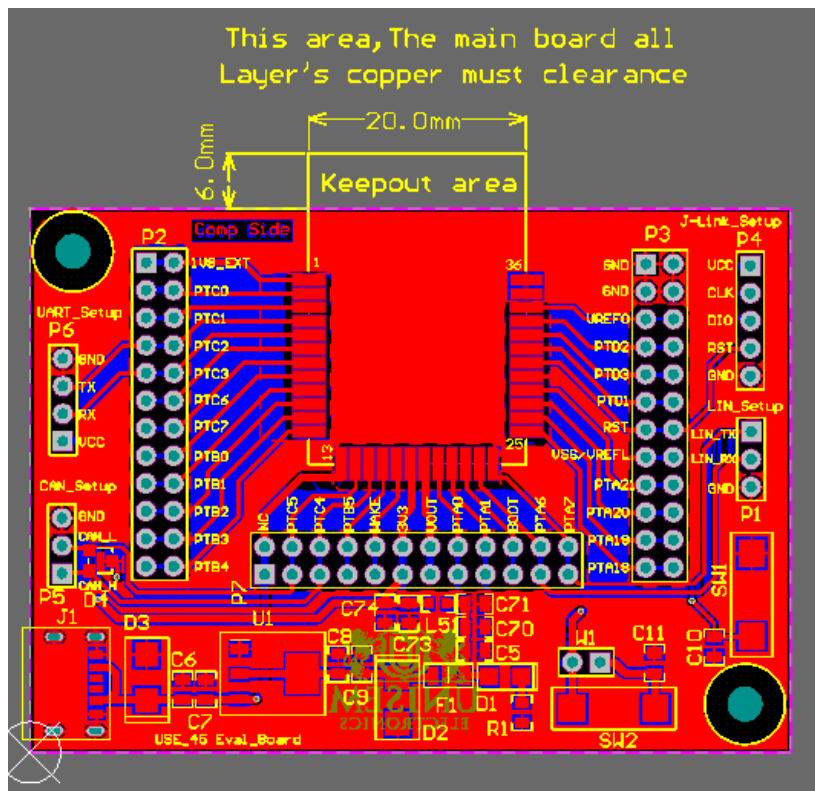


Fig 7



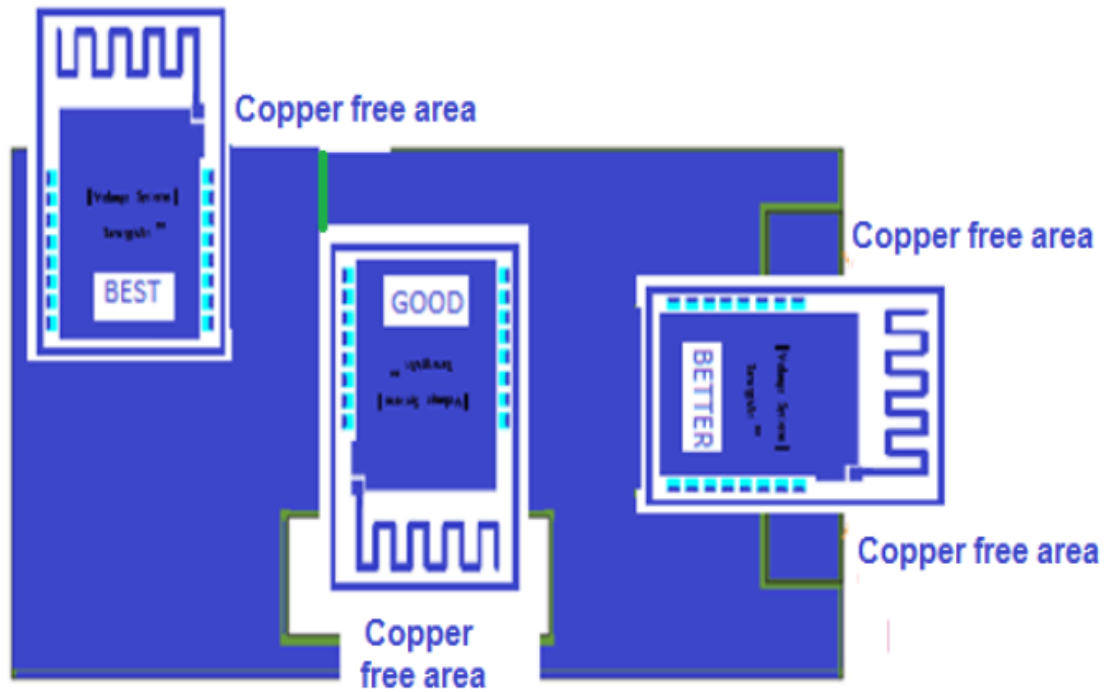
The module assembled in the motherboard, must pay attention to: Copper around the antenna and on all layers must be cleared. The module antenna should be on the edge, and there is no metal or obstruct electromagnetic radiation material to surround the antenna area.

Module placement strategy:

BLE/Sub-1GHz modules are RF devices that require proper placement on PCB to ensure optimal performance. The antenna on the PCB has (about) an omnidirectional radiation pattern. To maximize antenna efficiency, an adequate grounding plane must be provided under the module. However, the areas underneath and surrounding the antenna area must be free of copper. The position of the module on the host board and overall design of the product enclosure contribute to antenna performance. Poor design affects radiation patterns and can result in reflection, diffraction, and/or scattering of the transmitted signal thus limiting the range.

Basic guidelines:

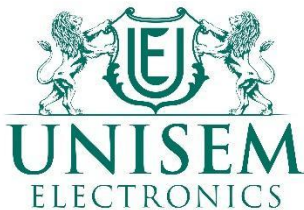
- i. Never place the ground plane or route copper traces directly underneath the antenna portion of the module.
- ii. Never place the antenna close to metallic objects.
- iii. Keep wiring, components and objects away from the antenna.
- iv. Do not place the antenna in a metallic or metalized plastic enclosure.
- v. Enclosure walls should be 1cm or more away from the antenna in all directions.
- vi. If possible, mount the antenna overhanging the edge of the host board (Motherboard). Add an uninterrupted ground plane on the host board, directly underneath the top shielded zone of the module, up to the PCB edge. Adding a ground plane will allow traces to be run on the bottom side of the host board if required.
- vii. If the antenna cannot be mounted in an overhanging position, then provisions must be made to keep the area clear of copper.
- viii. If any components containing metal conductor or conductive substance are placed close to the antenna, it might obstruct radio wave radiation, which causes a significant reduction in communication distance significantly.



Module Placement Recommendation

Fig 9

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